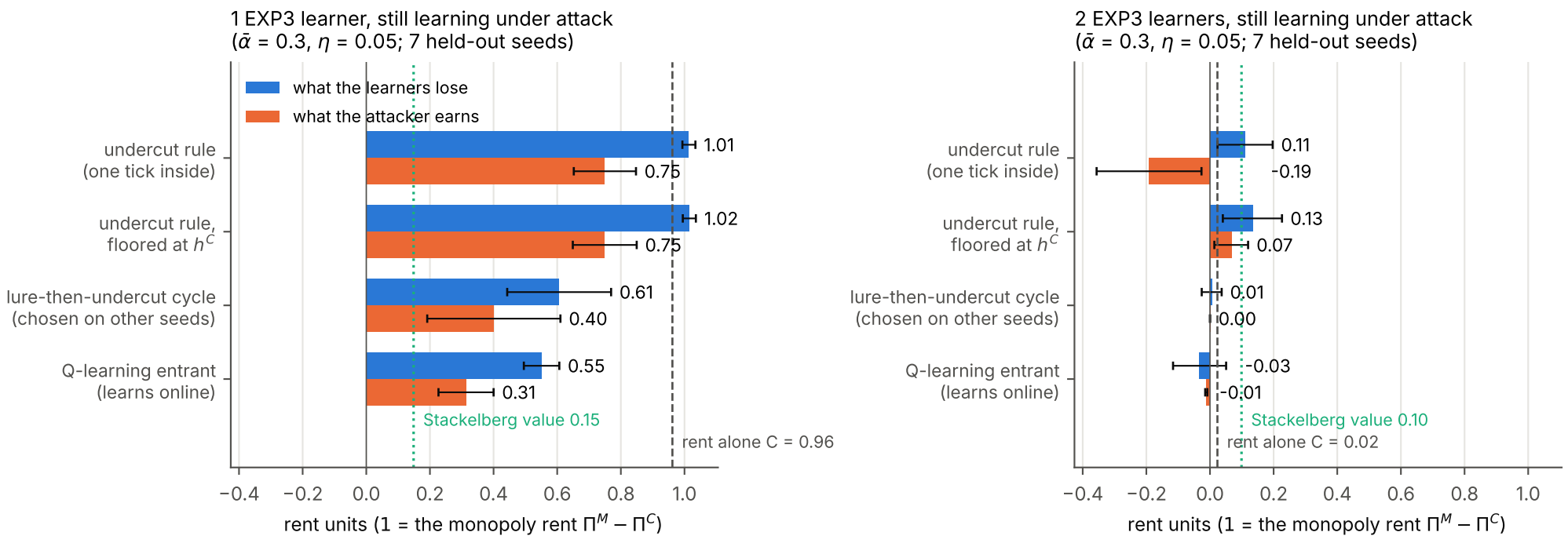
1 EXP3 learner, still learning under attack
($\bar{\alpha}$ = 0.3, $\eta$ = 0.05; 7 held-out seeds)
what the learners lose
what the attacker earns
undercut rule (one tick inside)
undercut rule, floored at $h^C$
lure-then-undercut cycle (chosen on other seeds)
Q-learning entrant (learns online)
1.01
0.75
1.02
0.75
0.61
0.40
0.55
0.31
Stackelberg value 0.15
rent alone C = 0.96
−0.4 −0.2 0.0 0.2 0.4 0.6 0.8 1.0
rent units (1 = the monopoly rent $\Pi^M - \Pi^C$)
2 EXP3 learners, still learning under attack
($\bar{\alpha}$ = 0.3, $\eta$ = 0.05; 7 held-out seeds)
undercut rule (one tick inside)
undercut rule, floored at $h^C$
lure-then-undercut cycle (chosen on other seeds)
Q-learning entrant (learns online)
0.11
-0.19
0.13
0.07
0.01
0.00
-0.03
-0.01
Stackelberg value 0.10
rent alone C = 0.02
−0.4 −0.2 0.0 0.2 0.4 0.6 0.8 1.0
rent units (1 = the monopoly rent $\Pi^M - \Pi^C$)